

Compact-operator models and unital consequences

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Scope

Faithfulness and both compact-image inclusions identify the nonunital compact-operator model in an unbundled sense. In the unital singleton case the identity becomes compact, yielding finite dimension and the exact full-operator image consequences.

12 direct Cards + 20 reused prerequisites = 32 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [A singleton model is faithful and exactly compact-valued](#)
- Unital consequence: [Finite-dimensional algebra in the unital singleton case](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

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Level 0 (9 Cards)

Level 0

Full operator image in finite dimension

Constructs a preimage of an arbitrary operator by separately lifting its self-adjoint real and imaginary parts.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.  
surjective_of_irreducible_of_finiteDimensional
```

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Singleton condition tested against all irreducible representations

Defines the unital singleton condition while allowing the quantified representations to be presented without a unit equation.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.  
IsSingletonIrreducibleModelAmongNonUnital
```

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#) · [Finite-dimensional algebra in the unital singleton case](#) · [Finite-dimensional representation space in the unital singleton case](#)

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Level 0

A compact-operator model for a representation

Packages faithfulness together with the equality $e(A) = \mathcal{K}(H)$.

```
MathlibAnnex.Analysis.CStarAlgebra.IsCompactOperatorModel
```

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#) · [Compact-operator conclusion for the unital singleton condition](#)

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Level 0

An isolated point in a countable Baire space

Applies the Baire property to the closed cover by singletons.

```
MathlibAnnex.Topology.exists_isOpen_singleton
```

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unital singleton model acts in finite dimension](#) · [An isolated character away from the scalar character](#)

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Level 0

Maximal abelian *-subalgebras

Defines maximality by inclusion among commutative unital *-subalgebras.

`MathlibAnnex.Analysis.CStarAlgebra.IsMaximalAbelian`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A unital singleton model acts in finite dimension](#) · [A maximal abelian subalgebra containing a self-adjoint element](#)

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Level 0

Pure states as real extreme points

Defines purity inside the normalized positive state space.

`MathlibAnnex.Analysis.CStarAlgebra.IsPureState`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

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Level 0

The preimage ideal of the compact operators

Defines the two-sided ideal $J_\rho = \rho^{-1}(\mathcal{K}(H))$.

`MathlibAnnex.CStarAlgebra.compactPreimageIdeal`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#) · [A unital singleton model acts in finite dimension](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

A character separating a closed ideal

Finds one character which kills a closed ideal and detects a specified element outside it.

`MathlibAnnex.Analysis.CStarAlgebra.exists_character_annihilating_closedIdeal_of_not_mem`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Every operator in a singleton image is compact](#)

[Open Declaration Card](#) [Read exact source](#)

Level 0

Representations without a unit-preservation requirement

Names *-representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

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Level 1 (6 Cards)

Level 1

The GNS representation of a pure state is irreducible

Applies the cyclic pure-state criterion to the space and vector constructed from the same state.

`MathlibAnnex.Analysis.CStarAlgebra.isIrreducible_pureState_gnsStarAlgHom`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A faithful singleton model forces simplicity](#) · [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

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Level 1

A character extends to a pure state

Chooses an extreme state in the compact face of all extensions of one prescribed character.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_extension`

Immediate Card prerequisites: [Pure states as real extreme points](#)

Used by in this scope: [A pure state detects a nonzero square](#) · [A character becomes a joint unit eigenvector](#)

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Level 1

Unitary equivalence of two representations

Defines equivalence by one surjective isometry intertwining all algebra actions.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

An isolated character away from the scalar character

Finds an isolated point in the open complement of the distinguished scalar character.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_isolated_character_ne_infinity`

Immediate Card prerequisites: [An isolated point in a countable Baire space](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

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Level 1

Nonzero irreducibility without a unit assumption

Requires nonzero action as well as the absence of proper closed reducing subspaces.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

A maximal abelian subalgebra containing a self-adjoint element

Starts with the algebra generated by the given element and extends it by inclusion-maximality.

`MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint`

Immediate Card prerequisites: [Maximal abelian *-subalgebras](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

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Level 2 (3 Cards)

Level 2

A representative of the unique irreducible class

Defines when one nonzero irreducible *-representation represents the sole unitary-equivalence class of such representations.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

Immediate Card prerequisites: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A character becomes a joint unit eigenvector](#)

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Level 2

A pure state detects a nonzero square

Detects a nonzero element through its positive square without an algebra separability assumption.

`MathlibAnnex.Analysis.CStarAlgebra.exists_pureState_nonzero_on_star_mul_self`

Immediate Card prerequisites: [A character extends to a pure state](#)

Used by in this scope: [Faithfulness from a unique irreducible class](#) · [A unital singleton model acts in finite dimension](#)

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Level 2

A faithful singleton model forces simplicity

Uses irreducible representations annihilating proper ideals to rule out nonzero proper closed ideals.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.isSimpleCStarAlgebra_of_singleton_of_injective`

Immediate Card prerequisites: [The GNS representation of a pure state is irreducible](#)

Used by in this scope: [A unital singleton model acts in finite dimension](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3 (3 Cards)

Level 3

Faithfulness from a unique irreducible class

Uses a pure state on the unitization to detect any hypothetical nonzero element of the kernel.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.injective_of_singleton`

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [A pure state detects a nonzero square](#) · [The GNS representation of a pure state is irreducible](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

[Open Declaration Card](#) [Read exact source](#)

Level 3

A unital singleton model acts in finite dimension

Uses simplicity to put the identity operator in the compact image ideal.

`MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton`

Immediate Card prerequisites: [Maximal abelian *-subalgebras](#) · [A faithful singleton model forces simplicity](#) · [A pure state detects a nonzero square](#) · [The preimage ideal of the compact operators](#) · [An isolated point in a countable Baire space](#)

Used by in this scope: [Compact-operator conclusion for the unital singleton condition](#) · [Finite-dimensional algebra in the unital singleton case](#) · [Finite-dimensional representation space in the unital singleton case](#)

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Level 3

A character becomes a joint unit eigenvector

Transfers a normalized GNS eigenvector to the given representation through a unitary with a specified direction.

```
MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_unit_eigenvector_of_character_ne_infinity
```

Immediate Card prerequisites: [A representative of the unique irreducible class](#) · [The GNS representation of a pure state is irreducible](#) · [A character extends to a pure state](#)

Used by in this scope: [The full character space is countable](#)

[Open Declaration Card](#) [Read exact source](#)

Level 4 (4 Cards)

Level 4

Compact-operator conclusion for the unital singleton condition

Applies the nonunital compact-operator theorem to the same operator-valued map underlying a unital representation.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.faithful_and_compactOperatorModel_of_singleton_amongNonUnital
```

Immediate Card prerequisites: [A compact-operator model for a representation](#) · [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#) · [Full operator image in finite dimension](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 4

Finite-dimensional representation space in the unital singleton case

Applies the unital finite-dimensional-space theorem after checking the representation-interface conversion.

```
MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_space_of_singleton_amongNonUnital
```

Immediate Card prerequisites: [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 4

Finite-dimensional algebra in the unital singleton case

Distinguishes finite dimension of the representation space from finite dimension of the algebra embedded in its operators.

MathlibAnnex.Analysis.CStarAlgebra.Representation.finiteDimensional_algebra_of_singleton_amongNonUnital

Immediate Card prerequisites: [Singleton condition tested against all irreducible representations](#) · [A unital singleton model acts in finite dimension](#)

Used by in this scope: [An infinite-dimensional unital algebra has no separable singleton model](#)

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Level 4

The full character space is countable

Uses separated unit eigenvectors to count the non-scalar characters, then restores the one scalar character.

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.countable_characterSpace_of_nonUnital_singleton

Immediate Card prerequisites: [A character becomes a joint unit eigenvector](#)

Used by in this scope: [A nonzero projection with scalar corner](#)

[Open Declaration Card](#) [Read exact source](#)

Level 5 (2 Cards)

Level 5

An infinite-dimensional unital algebra has no separable singleton model

Uses the finite-dimensional algebra conclusion to contradict the proposed singleton condition.

MathlibAnnex.Analysis.CStarAlgebra.Representation.not_singleton_amongNonUnital_of_infiniteDimensional

Immediate Card prerequisites: [Finite-dimensional algebra in the unital singleton case](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)

Level 5

A nonzero projection with scalar corner

Constructs a projection in the original algebra from an isolated non-scalar character of a unitization subalgebra.

MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_scalar_corner

Immediate Card prerequisites: [A maximal abelian subalgebra containing a self-adjoint element](#) · [An isolated character away from the scalar character](#) · [The full character space is countable](#)

Used by in this scope: [A projection represented by a rank-one operator](#)

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Level 6 (1 Card)

Level 6

A projection represented by a rank-one operator

Obtains all projection, normalization, image and compactness conditions on the same witnesses.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_nonzero_projection_rankOne_map`

Immediate Card prerequisites: [Faithfulness from a unique irreducible class](#) · [A nonzero projection with scalar corner](#)

Used by in this scope: [Every rank-one operator has an algebra preimage](#)

[Open Declaration Card](#) [Read exact source](#)

Level 7 (1 Card)

Level 7

Every rank-one operator has an algebra preimage

Turns one represented rank-one projection into every rank-one operator using dense orbits and a closed range.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_rankOne_of_singleton`

Immediate Card prerequisites: [A projection represented by a rank-one operator](#)

Used by in this scope: [Compact operators lie in a singleton representation range](#) · [Every operator in a singleton image is compact](#)

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Level 8 (2 Cards)

Level 8

Every operator in a singleton image is compact

Excludes a noncompact image by making one separating character take both zero and one on the same projection.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.isCompactOperator_map_of_singleton`

Immediate Card prerequisites: [Every rank-one operator has an algebra preimage](#) · [A character separating a closed ideal](#) · [The preimage ideal of the compact operators](#)

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#)

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Level 8

Compact operators lie in a singleton representation range

Obtains the inclusion of all compact operators in the image of the original algebra.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.exists_preimage_of_compact_singleton`

Immediate Card prerequisites: [Every rank-one operator has an algebra preimage](#)

Used by in this scope: [A singleton model is faithful and exactly compact-valued](#)

[Open Declaration Card](#) [Read exact source](#)

Level 9 (1 Card)

Level 9

A singleton model is faithful and exactly compact-valued

Combines faithfulness with both inclusions needed to identify the represented range.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.faithful_and_compactOperatorModel_of_singleton`

Immediate Card prerequisites: [Compact operators lie in a singleton representation range](#) · [A compact-operator model for a representation](#) · [Every operator in a singleton image is compact](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)